

# Data Science

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## Lecture 07: Data Exploration and Preprocessing

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Converted from LNKB-tex part-en-ds-chp3-data.tex (two source units re-encoded from ISO-8859). Uses Lecture 02's quantiles and covariance throughout.



# Module 1

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## What Kind of Data Is This?

**Guiding Question:** *The file loads and the numbers compute. Why is that not yet a reason to trust the average?*

# Four Attribute Types

## Nominal

*categorical,  
qualitative*

Labels only: eye  
colour, ZIP code, ID.

Meaningful:  $=$ ,  $\neq$

## Ordinal

*categorical,  
qualitative*

Ordered labels:  
grades, rankings,  
"small/medium/large".

Meaningful:  $<$ ,  $>$

## Interval

*numeric, quantitative*

Differences are  
meaningful, the zero  
point is arbitrary:  
temperature in  $^{\circ}\text{C}$ ,  
calendar dates.

Meaningful:  $+$ ,  $-$

## Ratio

*numeric, quantitative*

A true zero, so ratios  
are meaningful:  
length, mass, counts,  
duration.

Meaningful:  $\times$ ,  $\div$

The four types form a hierarchy: each level permits everything the previous one does, plus more. The type is a property of the *measurement*, not of the storage format — which is why a ZIP code stored as an integer is still nominal.

# Why the Type Decides the Method

## The classic error

Averaging a nominal attribute. The mean of ZIP codes 10115 and 81667 is a number, and it is meaningless.

## The subtle one

20 °C is not "twice as warm" as 10 °C — interval scales have no true zero, so ratios do not transfer.

The attribute type determines which **statistics**, which **distance measures** and which **plots** are permissible. Every later chapter depends on it.

# Mini-Exercise

Classify each attribute, and name one operation that would be invalid:

1. Customer satisfaction on a five-point scale
2. Year of birth
3. Annual income
4. Preferred payment method

# Mini-Exercise — Solution

An attribute's **scale of measurement** licenses the operations you may perform on it. Choosing a statistic before checking the scale is the most common silent error in data analysis.

# Mini-Exercise — Solution

**1. Ordinal.** Averaging is questionable — the gaps between points are not known to be equal.

**2. Interval.** "Twice the year" is meaningless differences (age) are fine.

**3. Ratio.** All operations valid zero income is a real zero.

**4. Nominal.** Only counting and comparison for equality.

An attribute's **scale of measurement** licenses the operations you may perform on it. Choosing a statistic before checking the scale is the most common silent error in data analysis.

# Four Shapes of Data

## Record data

Objects × attributes  
— the familiar table.  
Rows are  
independent of each  
other.

## Graph data

Objects carry **links** to  
other objects: web  
pages, social  
networks, citations.

## Fields / grids

Objects sit in a  
**spatial** arrangement:  
images, sensor  
lattices, simulations.

## Spatio-temporal

Position **and** time:  
monthly land  
temperatures, traffic  
movement,  
trajectories.

The shape decides which methods even apply. A table can be shuffled without loss a grid or a time series cannot — the arrangement *is* information.

Ask what is lost by flattening each non-record type into a table. That question is the bridge to network science and to time-series methods, and it recurs whenever someone loads a CSV.

# Three Data Quality Problems

## Noise

Random distortion of measured values. Cannot be removed, only modelled or smoothed.

## Outliers

Values far from the rest.  
Either an **error** to remove or the **signal** you were looking for — fraud, faults, discoveries.

## Missing values

Absent entries. Handling depends on **why** they are missing, which the data itself does not tell you.

"Outlier" is not a property of a data point; it is a judgement about a model of what is normal. Deleting outliers deletes evidence against that model.

# Mini-Exercise

A sensor dataset has 8% missing temperature readings.

1. Give two mechanisms that could produce them.
2. For each, is deleting those rows acceptable?
3. What would you need to know to decide?

# Mini-Exercise — Solution

The shape of the data decides which methods apply the **reason** for a quality problem decides how it may be handled. Neither is visible in the file.

This is Principle 3 in its most practical form: a preprocessing decision is a modelling assumption, and here an untestable one. Students should leave able to ask "why is this missing?" before "how do I impute it?".

# Mini-Exercise — Solution

1. Random transmission dropouts — or the sensor failing **when it is very cold**.

2. Dropouts: deletion is roughly harmless. Cold-failure: deletion silently removes the coldest days and biases every statistic afterwards.

3. The **mechanism** — whether missingness depends on the unobserved value itself. The data alone cannot answer this domain knowledge must.

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# Module 2

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## Seeing the Data

**Guiding Question:** *Summary statistics answer the questions you thought to ask. What shows you the ones you did not?*

# Visualization Is Everywhere

Newspaper tables, subway maps, weather charts, stock graphs, a 3-D reconstruction of an injured knee, demographic bar charts, social-network layouts.

**What do these have in common?** All of them present data in a form the eye processes faster than the equivalent numbers.

Question-first (P1). Let students name the common property before it is stated; "you can see it at a glance" is the answer, and the next slide gives the reason.



# Why It Works

## Bandwidth

Vision is the human sense with the highest data rate — pattern detection runs in parallel and pre-attentively.

## Interaction

A picture is a natural surface for exploring: zoom, filter, select, compare.

## Discovery

Visualization surfaces the **unexpected** — the structure nobody thought to compute a statistic for.

Summary statistics answer questions you already asked. A plot lets the data ask you one — which is exactly what Anscombe's quartet demonstrated.

Visualization is not decoration for results; it is an **instrument of analysis**, and the only one that can show you what you did not think to measure.

Deliberate callback to L01 and L02 (spiral): the same argument, now with the tools to act on it. This unit is the hinge between "data does not speak for itself" and the concrete plot types that follow.



# The Boxplot

A visual summary of a distribution built from quartiles:

## The box

Spans  $Q_1$  to  $Q_3$  — the interquartile range, the middle half of the data.

## The line

The **median** inside the box. Its position shows skew.

## The whiskers

Minimum and maximum, or (Tukey) the most extreme points within  $1.5 \times \text{IQR}$ . Points beyond them are drawn as **outliers**.

The  $1.5 \times \text{IQR}$  rule is a **convention**, not a test. It flags candidates for inspection, not errors for deletion.

## Speaker notes

Boxplots are built entirely from the quantiles the students defined in Lecture 02 — say so. Their strength is comparison: ten boxplots side by side are readable where ten histograms are not.



# The Histogram

## What it shows

The **shape** of a distribution: modality, skew, gaps and clusters that a boxplot hides entirely.

## What it hides

Everything depends on the **bin width**. Too wide and structure vanishes too narrow and noise looks like structure.

A bimodal distribution and a unimodal one can produce the same boxplot. Always look at one histogram before trusting a five-number summary.

# Mini-Exercise

Two datasets have identical minimum,  $Q_1$ , median,  $Q_3$  and maximum.

1. Must they have the same histogram?
2. Sketch two distributions with the same boxplot but a visibly different shape.
3. What does this repeat from an earlier lecture?

# Mini-Exercise — Solution

A boxplot compares distributions compactly; a histogram shows their shape. **Neither alone is sufficient**, and the boxplot's outlier rule is a convention you may overrule.

# Mini-Exercise — Solution

**1.** No. Quartiles fix five points the mass between them can be arranged freely.

**2.** A uniform spread versus two tight clusters at the ends of the box — same quartiles, opposite stories.

**3.** Anscombe's quartet: summaries agree, the data does not. The lesson is scale-free.

A boxplot compares distributions compactly; a histogram shows their shape. **Neither alone is sufficient**, and the boxplot's outlier rule is a convention you may overrule.

# The Scatterplot

An X–Y plot of two dimensions against each other — the standard instrument for inspecting a **relationship** rather than a distribution.

With more than two dimensions there are four options:

## Subsetting

Let the user choose which dimensions to display.

## Reduction

Project to fewer dimensions (see preprocessing, and projections later).

## Embedding

Map extra dimensions to other channels: colour, size, shape.

## Multiple displays

Superimpose, or juxtapose in a **scatterplot matrix**.

The four options are exhaustive and worth memorising — every high-dimensional visualization technique is one of them or a combination. Colour and size are channels with far lower precision than position use them for secondary variables.

# What a Scatterplot Reveals

Linear association, non-linear structure, clusters, outliers and heteroscedasticity — most of which no single correlation coefficient reports.

A correlation of 0 is consistent with a perfect parabola. The scatterplot shows it the coefficient does not.

# Tufte's Principle

**Maximise the data-ink ratio:** every drop of ink on the page should carry information. Delete anything that does not — heavy gridlines, boxes, redundant tick marks, decorative fills.

**Range frames** apply the principle to the axes: draw each axis only over the interval the data actually spans, so the frame itself reports the minimum and maximum.

The exercise: take a finished chart and ask of each element, *what would I lose by deleting this?*

A scatterplot shows the structure a coefficient summarises away. Beyond two dimensions you must **subset, reduce, embed or juxtapose** — and every mark you draw should earn its ink.

Tufte's rule is a good design heuristic and an over-applied dogma note that some redundancy aids reading under time pressure. The point students should keep is intentionality: each element defended, not each element removed.

# When a Line Is Justified

Connecting points with a line asserts an **order** between them — and often an interpolation.

**Data requirement:** one or more quantitative value attributes on a comparable scale, plus one **ordered** key attribute (usually time).

Connecting nominal categories with a line claims an order that does not exist. The line is a statement, not a decoration.

This is the attribute-types unit paying off immediately: the permissible plot follows from the scale of the key attribute. Bar charts for nominal, lines for ordered.

# Aspect Ratio Is Not Cosmetic

Humans judge **angles** far more accurately than absolute slopes, so the shape of the plotting area changes what the reader concludes. The same series looks flat in a wide frame and dramatic in a tall one.

**Banking to 45°** chooses the aspect ratio so that the average line segment sits near 45 degrees — the angle at which slope differences are most discriminable.

## Additional channels

Colour and line type distinguish series — but only for a handful of them.

# Multiple Series

## Overlaid

Best for comparing a few series directly against each other.

## Stacked

Each series is drawn on top of the previous one. Shows the total well — and makes individual series **hard to read**, since each sits on a moving baseline.

## Sorted / small multiples

One panel per series, ordered by some property. Scales to many series where overlaying fails.

A line asserts order and interpolation the **aspect ratio** decides what the reader sees and stacking trades the readability of the parts for the readability of the whole.

The stacked-plot critique is worth demonstrating: ask students to read the third band's trend off a stacked chart — they cannot, because the baseline moves. Small multiples are the usual better answer.

# Module 3

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## Preprocessing

***Guiding Question:*** *Most of a project is spent before the model. What is actually being decided in that time?*

# Aggregation

Combine several objects or attributes into one — customer purchases summarised by store, by month, by price band.

## Gains

Fewer objects to process **less variation**, hence more stable estimates the natural operation of a data warehouse.

## Costs

Detail is gone, irreversibly. Effects that exist only within a group become invisible.

Speaker notes

Aggregation is where Simpson's paradox lives: a trend can reverse when groups are pooled. Worth one sentence — students will meet it in dashboards long before they meet it in a statistics course.



# Sampling

Work with a subset of the objects: random sampling with or without replacement, or **stratified** sampling that preserves group proportions.

Sampling always loses information. The question is whether it loses anything that matters — a sufficiently large sample retains the overall structure.

The sine-wave example:  
sample too sparsely and  
the reconstructed signal is  
not merely rougher, it is  
**wrong**.

# Dimensionality Reduction

**The curse of dimensionality.** As the number of attributes grows, the data becomes sparse: distances between points concentrate, neighbourhoods lose meaning, and the volume to be covered grows exponentially.

## The problem

Many features are irrelevant, noisy or correlated.

## The cost

More computation, more parameters, more variance — and weaker distance-based methods.

## The response

**Feature selection** keeps a subset **feature extraction** builds new combined dimensions.

Every preprocessing step **trades information for tractability**. The trade is usually worth it — but it is a modelling decision, and it should be recorded as one.

Point forward: projections (PCA, MDS) are the formal treatment of feature extraction, and clustering suffers directly from the curse. This unit is the motivation both later lectures rely on.

